

Zihan Ruan

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Research Interests: LLM Serving, KV Cache Management, AI Infrastructure

Education

University of British Columbia

Starting Sep. 2026

Master of Engineering in Electrical Engineering, Incoming Student

Fuzhou University – Maynooth University Joint Program

Sep. 2022 – Jun. 2026

B.Eng. in Electronic Information Engineering, Fuzhou University

B.Eng. (Honours) in Electronic Engineering, Maynooth University

First-Class Honours • Cumulative GPA: 3.41/4.00 • Recent 60-Credit GPA: 3.86/4.00

Selected Coursework: Computer and Communication Networks(4.0), Software Engineering(4.0), Object-Oriented Programming(4.0), Computer Control Systems(4.0), and Embedded Real-Time Systems(3.7)

Research Experience

Long-CoT LLM Serving and KV Cache Management

Apr. 2026 – Present

Research Intern, AIMNet Lab, East China University of Science and Technology

- Identified and characterized a **generated-KV-dominant Long-CoT serving regime**, in which long reasoning outputs cause generated KV states to dominate memory consumption despite relatively short input prompts.
- Designed and executed a reproducible motivation study covering more than **16.9 million requests** across reasoning datasets and production serving traces. Introduced the **Generated KV Ratio** and showed that generated KV accounts for approximately **96–98%** of the total KV cache in representative Long-CoT workloads, limiting the median memory-reduction potential of prompt-only optimization to approximately **2–4%**.
- Constructed a **20-stage end-to-end lifecycle model** covering request progression, token and KV state transitions, and runtime decisions from generation and observation to retention, compression, placement, migration, and reclamation.
- Applied the lifecycle model to systematically compare existing KV cache methods by their observable states, decision points, management actions, optimization objectives, and system overheads, exposing gaps in online generated-KV management and cross-stage coordination.
- Currently investigating algorithmic mechanisms for state-aware generated-KV retention and compression, together with system policies for GPU-memory-aware placement, migration, hierarchical storage, and concurrent multi-request scheduling.

Publications

1. T. Shang, Z. Li, P. Xu, J. Qiao, G. Chen, **Z. Ruan**, and W. Hu, “Bridging Text and Vision: A Multi-View Text-Vision Registration Approach for Cross-Modal Place Recognition,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025. **Oral Presentation.**
2. **Z. Ruan**, “Reasons and Solutions for Device Mismatch and Structural Mismatch in Electronic Circuits,” *Applied and Computational Engineering*, vol. 101, 2024, Art. no. 20241045. DOI: [10.54254/2755-2721/101/20241045](https://doi.org/10.54254/2755-2721/101/20241045).

Teaching Experience

Fuzhou University

Sep. 2025 – Feb. 2026

Teaching Assistant, EE302 Real-Time and Embedded Systems

- Mentored **15 students** on embedded-systems concepts, programming assignments, and laboratory troubleshooting.
- Co-administered laboratory assessments for approximately **150 students**, including rubric-based grading, result verification, and report evaluation.

- Provided feedback on code quality, experimental methodology, data analysis, and technical writing.

Selected Projects

Low-Light Image Enhancement and 2× Super-Resolution Sep. 2025 – Apr. 2026

Undergraduate Thesis Project

- Designed and implemented a two-stage image restoration pipeline combining low-light enhancement and 2× super-resolution on the LSRW-Huawei dataset.
- Integrated DMFourLLIE and MambaIRv2 into a unified **PyTorch** workflow covering data preprocessing, model training, inference, and performance evaluation.
- Fine-tuned and retrained MambaIRv2 on low-light-enhanced images produced by DMFourLLIE, adapting the super-resolution model to residual noise, artifacts, and distribution shifts introduced by the enhancement stage.
- Improved image restoration performance over the original super-resolution model, achieving gains in both **PSNR** and **SSIM** through pipeline-specific fine-tuning.

Image Inpainting Aug. 2024 – Jan. 2025

- Adapted a PyTorch-based image inpainting pipeline for preprocessed custom datasets and integrated pretrained model weights from Hugging Face.
- Executed layer-unfreezing and retraining experiments and analyzed restoration performance across foreground and background regions.

Industry Experience

Jingwei Hirain R&D Center, Tianjin Feb. 2025 – Jun. 2025

Radar Testing Engineer, Part-Time

- Designed and executed validation cases for automotive front and corner radar systems, covering test-environment setup, data collection, result verification, and failure analysis.
- Developed a **C++ based CAN gateway** to forward vehicle-body signals to the radar test host, enabling synchronized acquisition and analysis of radar and vehicle-state data.
- Parsed CAN messages using DBC files and analyzed real-time signal traces to verify radar behavior and diagnose communication and test failures.
- Consolidated test results and failure analyses into engineering reports and provided recommendations for product and test-process improvement.

State Grid Xintong Yili Technology Co., Ltd., Fuzhou Branch Jul. 2024 – Aug. 2024

Assistant Algorithm Engineer, Part-Time

- Implemented introductory computer-vision experiments using PyTorch and OpenCV, including image preprocessing and handwritten-digit recognition.
- Prepared technical documentation, experiment records, testing materials, and internal presentation slides.

Honors and Awards

- **Comprehensive Scholarship (Third Class)**, Fuzhou University Maynooth International Engineering College, 2025–2026.
- **Honorable Mention**, The Interdisciplinary Contest in Modeling(ICM), 2024.